Pitch Sequence Prediction: MLB → Polish Ekstraliga Transfer Learning

Quoc Dat Nguyen | MSDS | BU

Motivation

During the 2023 World Baseball Classic, a controversial pitching substitution sparked a question: can machine learning help managers make better in-game decisions? This project builds a pitch outcome prediction model trained on MLB data and adapts it to the Polish Ekstraliga — a small amateur league with far less data — using transfer learning.

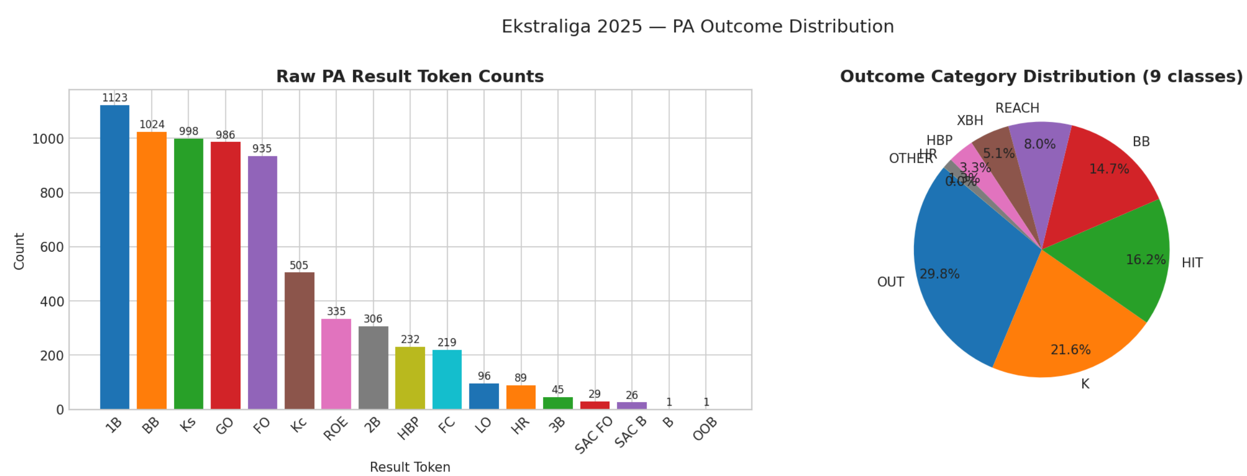
What Was Built

Three things: a pitch outcome predictor, a transfer learning pipeline, and a live in-game decision support tool that ranks pinch hitters and relief pitchers by expected value.

Exploratory Data Analysis

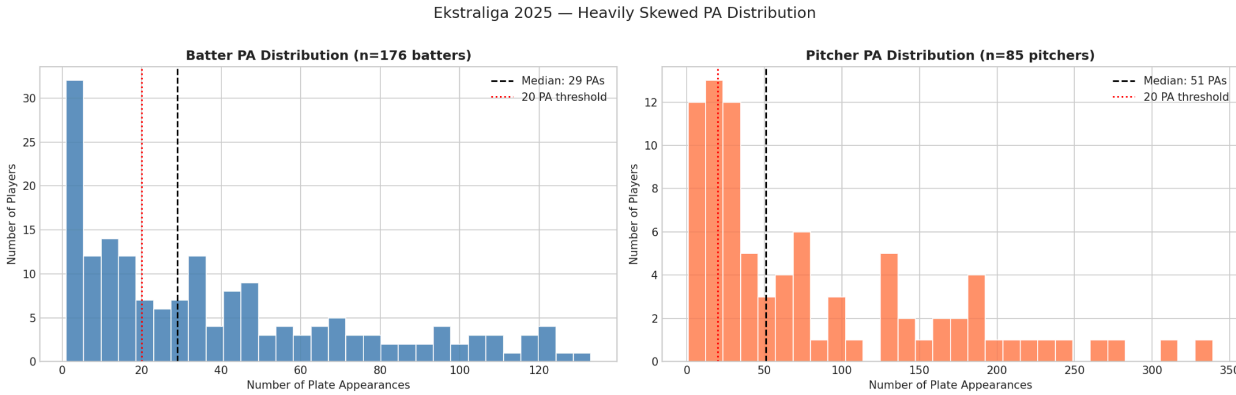
Before any modelling, the dataset was analysed to understand its structure, quality, and limitations. These findings directly shaped every modelling decision.

Outcome Distribution



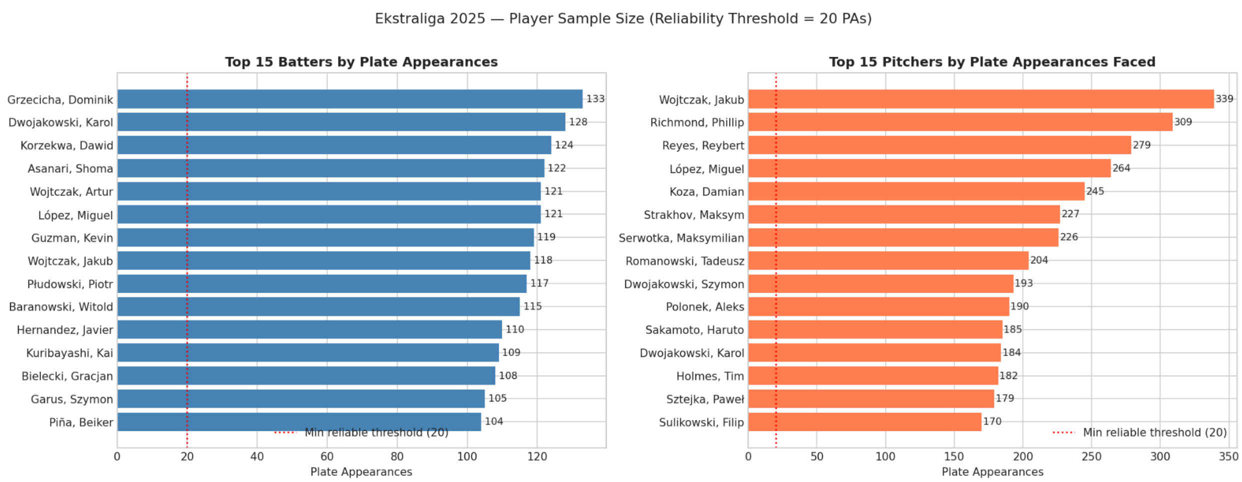
The dataset contains 9 outcome categories. OUT is the most common at roughly 30% of plate appearances, while OTHER appears only once. The class imbalance ratio between the most and least common outcomes exceeds 2,000 to 1. This imbalance is why class-weighted loss functions are used during LSTM training — without weighting, the model would learn to predict OUT for everything and achieve a misleading 30% accuracy.

The Core Data Problem: Skewed Plate Appearance Counts



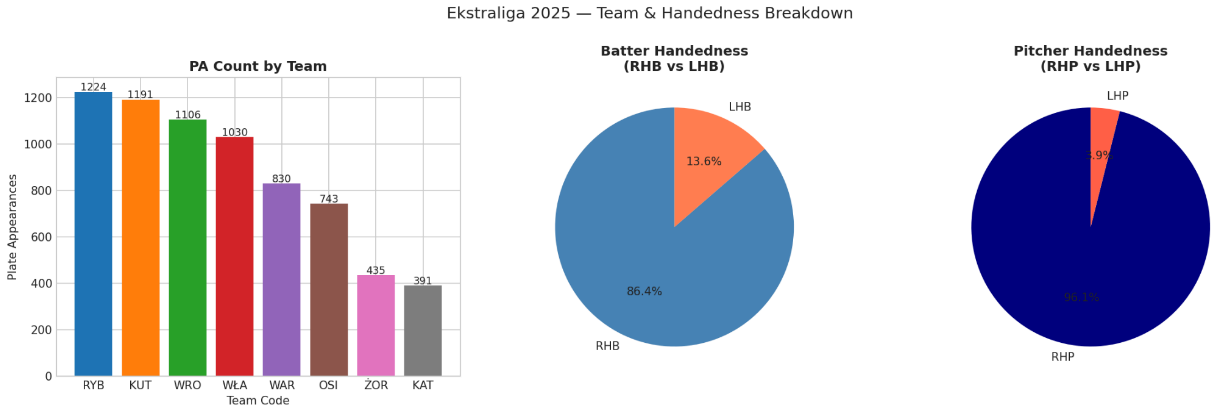
This is the single most important finding from the EDA. The dataset has 176 unique batters and 85 unique pitchers, but their plate appearance counts are extremely skewed. The median batter has only 29.5 plate appearances, the 25th percentile is just 9.75, and over half of all players appear fewer than 20 times. Some players appear only once.

This directly explains why the Markov model — which ignores player identity entirely — outperforms the LSTM. Tendency features like K rate, BB rate, and swing rate computed from 5 to 10 plate appearances are noisy estimates, not reliable statistics. The LSTM is trying to weight unreliable numbers.



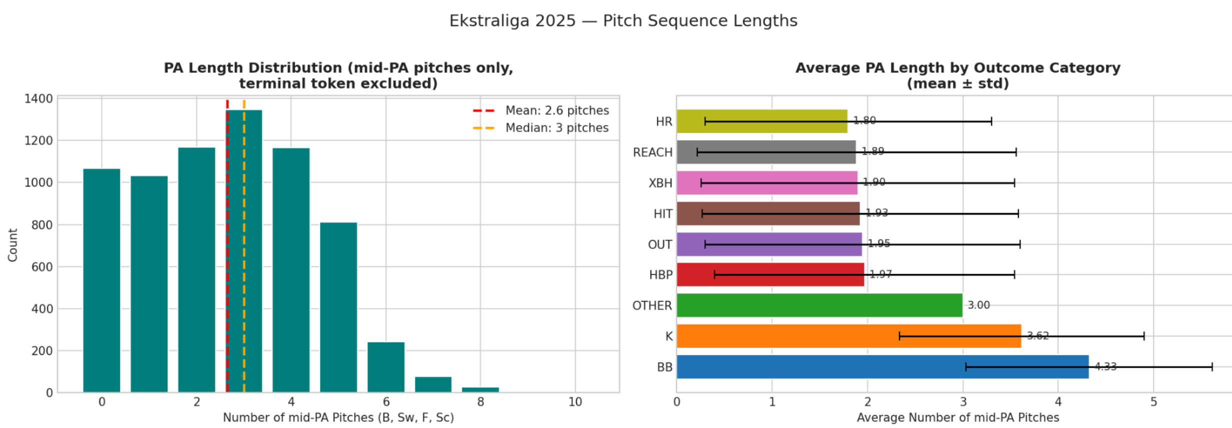
The top 15 batters and pitchers by sample size illustrate the gap: Grzecicha, Dominik leads with 133 plate appearances while Rivera, Yasnier has just 2. The red dotted line at 20 PAs marks the minimum threshold for statistically meaningful tendency features. Most players fall below it.

Team and Handedness Distribution



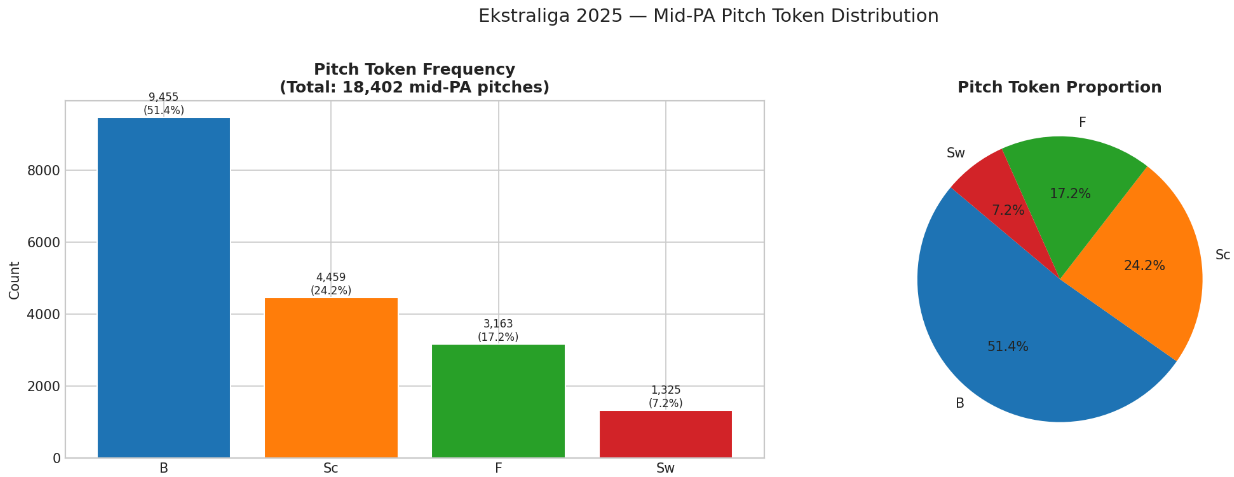
PA counts vary across the 8 teams in the dataset, with RYB contributing the most appearances and KAT the fewest. More critically, 86% of batters are right-handed (RHB) and 96% of pitchers are right-handed (RHP). This extreme imbalance means the model has almost no training signal for platoon advantage situations involving left-handed pitchers — one of the most actionable factors in real substitution decisions.

Pitch Sequence Lengths



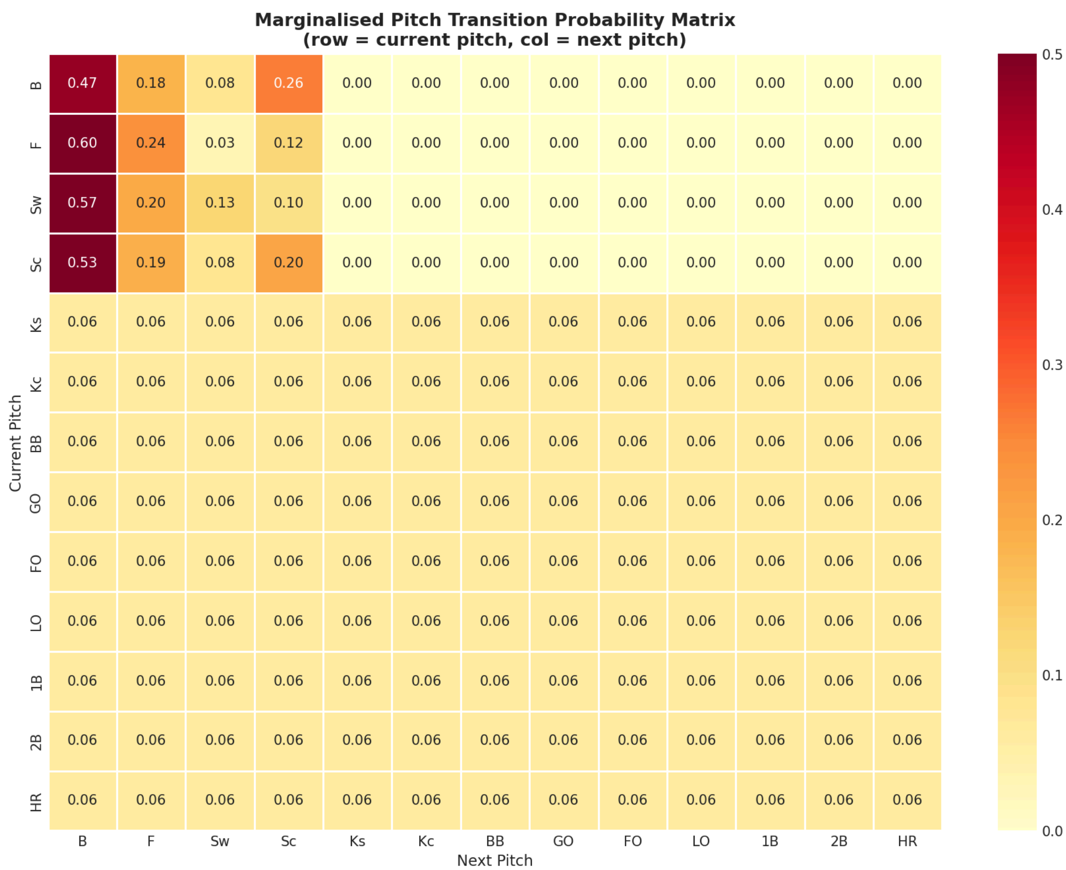
After stripping the terminal token from each sequence, the average plate appearance has only 1.5 mid-PA pitches. Many PAs end on the very first pitch, meaning the LSTM receives an empty sequence and must rely entirely on tendency features. The right panel shows that walks and strikeouts tend to have longer sequences while outs in play are shorter — this makes intuitive sense, as putting the ball in play quickly ends the at-bat.

Pitch Token Frequencies



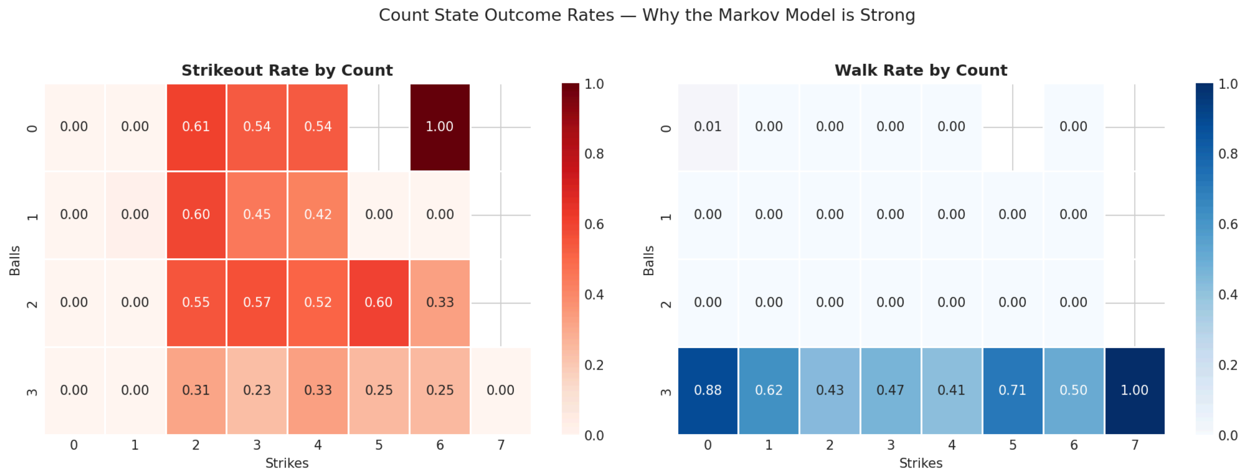
Ball (B) is the most common mid-PA token, followed by Foul (F) and Swinging Strike (Sw). Called Strike (Sc) is relatively rare. This reflects the reality of amateur baseball — pitchers throw more balls relative to called strikes than in MLB, and batters swing more frequently.

Markov Transition Matrix



This heatmap shows the probability of transitioning from one pitch token to the next, marginalised across all count states. Key patterns: after a Ball, another Ball is most likely. After a Swinging Strike, a strikeout becomes highly probable. After a Foul ball, another Foul or a Swinging Strike is most common. These transitions form the learned representations that the LSTM must discover from scratch, while the Markov model exploits them directly.

Why Count State Is So Predictive



This is why the Markov baseline is hard to beat. The K rate spikes dramatically at any 2-strike count — above 40% on 1-2 and 2-2 counts. The walk rate spikes just as sharply at 3-ball counts. The final count state when a plate appearance ends compresses enormous information about what just happened, and the Markov model reads it perfectly. Any neural network must learn this signal in addition to everything else.

Data

MLB: 33,095 plate appearances pulled from the MLB Stats API (2024 season). Features include pitch sequences, handedness, and count state.

Ekstraliga: 6,950 plate appearances from the 2025 season. Only pitch-level tokens are tracked — no velocity or spin rate like MLB. Sequences are shorter and data is far sparser.

The core challenge: MLB sequences are longer, physically richer, and structurally more consistent. Ekstraliga sequences are noisier, shorter, and spread across many players with very few appearances each.

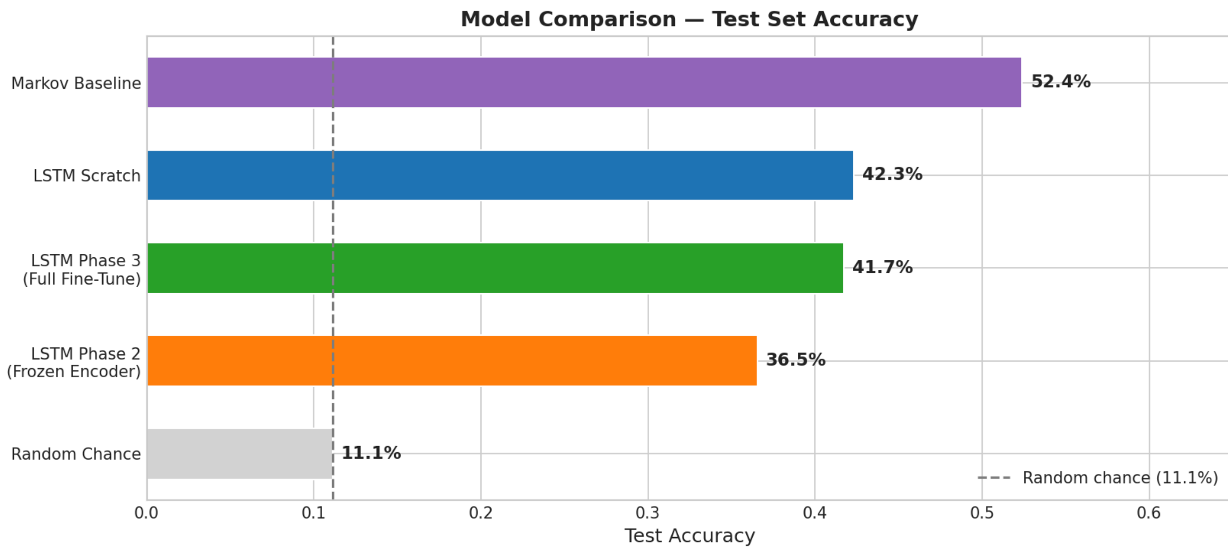
Models Tested

Markov Chain — predicts outcome purely from the final count state. Simple but surprisingly strong because count state is highly predictive.

LSTM from scratch — trained only on Ekstraliga data. Reads the pitch sequence plus player tendency features to predict outcome.

LSTM with transfer learning — pretrained on 33,095 MLB plate appearances, then fine-tuned on Ekstraliga in two phases: first with the encoder frozen, then with all layers unfrozen.

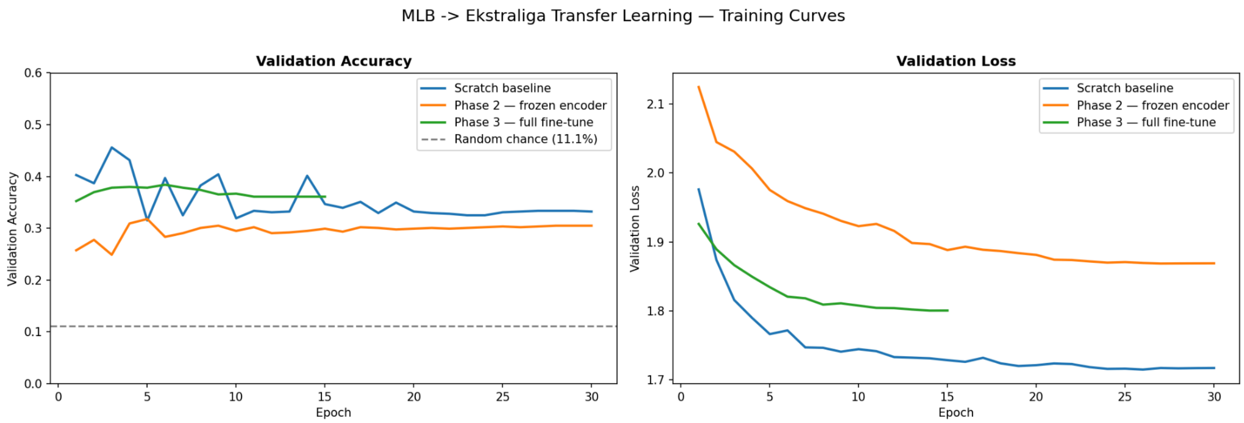
Results



Model	Test Accuracy
Random chance	11.1%
LSTM Phase 2 - frozen encoder	36.6%
LSTM Phase 3 - full fine tuning	41.7%
LSTM from scratch	42.3%
Markov Baseline	52.4%

All models beat random chance by 3–4×. The Markov model wins despite being the simplest — count state alone is highly predictive in a small dataset. The LSTM models are competitive but do not surpass Markov, which is an honest finding given only 6,950 training examples spread across 176 players most of whom have fewer than 20 appearances each.

Transfer Learning Findings



Freezing the MLB-pretrained encoder (Phase 2) hurts accuracy. MLB sequences are longer and physically richer — the frozen representations do not map cleanly onto short amateur sequences. Full fine-tuning (Phase 3) recovers most of the gap with scratch performance, and the loss curve shows it converges faster in early epochs. That faster convergence is the real transfer learning signal: the MLB-pretrained model starts from a better initialisation than random weights, even if the final accuracy ends up similar.

Decision Support Tool

The tool takes a live game situation — current pitcher, partial pitch sequence, inning, outs, runners — and ranks substitution candidates by expected value using run-expectancy weights.

Example: Bottom of the 7th, 1 out, runner on 1st. Sequence so far: Ball → Swing.

```
──────────────────────────────────────────────
  MATCHUP : Wojtczak, Jakub (RHB) vs Richmond, Phillip (RHP)
  Context : Inning 7 | 1 out(s) | Runners: 1--
  Sequence: B → Sw
──────────────────────────────────────────────
  Outcome     Probability  Distribution
  ───────     ───────────  ───────────
  K                  6.1%
  BB                 4.0%
  HBP                3.3%
  OUT               21.3%
  REACH             26.4%
  HIT               20.3%
  XBH               10.5%
  HR                 8.1%
  OTHER              0.0%
──────────────────────────────────────────────
  Batter EV  : +0.2881  (favours batter)
  Pitcher EV : -0.2881
──────────────────────────────────────────────
```

Karol Dwojakowski ranks #1, consistent with his season stats (.347 AVG, .484 OBP, lowest K rate of the three). The model gets the ranking direction right after cumulative season statistics were used as input features rather than noisy PA-log estimates.

Limitations

No game situation in training data. The Ekstraliga PA Logs do not record inning, score differential, or baserunner state per plate appearance. The decision support tool accepts these inputs and displays them, but the model itself ignores them — the predicted probabilities are identical whether it is the 1st inning or the 9th.

Small dataset. 6,950 plate appearances is not much for a neural network with 227,000 parameters. As the EDA shows, most players have fewer than 20 appearances, making reliable tendency features impossible to compute from the PA logs alone.

EV gaps between candidates are small. When candidates face the same partial sequence, the LSTM produces an identical hidden state for all of them. Differentiation comes only from the tendency MLP, which compresses player differences into a small 32-dimensional vector.

Handedness is hardcoded. The decision support tool defaults every player to RHB and RHP at inference, missing platoon advantages — one of the most actionable reasons to substitute in real baseball.

Future Work

Track game situation. Inning, outs, score, and runner state per plate appearance would make the substitution tool genuinely leverage-aware.

More data. Adding 2023 and 2024 seasons would triple the training set and likely close the gap with the Markov baseline.

Transformer architecture. Better suited to short sequences than LSTM. The BERT-style pretraining approach — treating pitch tokens like words — is the natural next step.

Better player embeddings. Learning a dense player embedding from all historical plate appearances would capture matchup dynamics that summary statistics miss.